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1

Mini Project (CSE7102) REVIEW-2 PRESENTATION

Date: 26-09-2026



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School of Computer Science and Engineering

CSS7102- Mini Project Review-2

Development of a Software Application for Analysis and Processing of DVB-S2 Receiver Output Stream

Project ID: PRJ_111 | Course: CSE7102
- Mini Project | Review-2 (26-09-2026)

Panel No: Panel 11 / PRJ_111

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PROBLEM STATEMENT

Format Heterogeneity: DVB-S2 receiver output streams exist across multiple alternative encapsulation layers (Baseband Frames, GSE packets, and MPEG Transport Streams), requiring fragmented, proprietary command-line utilities for inspection.

Lack of Intelligent Anomaly Detection: Conventional stream analyzers rely on basic static threshold checks, failing to detect subtle temporal anomalies, sudden jitter variations, corrupted modulation frames, or unexpected multiplex behaviors.

Absence of Unified Diagnostic Platforms: Existing open-source tools typically focus only on one isolated format (e.g., MPEG-TS only or PCAP inspection only) without providing unified health metrics, explainable AI diagnostics, comparative stream analysis, or automated report generation.

Need for Unified Software Workflow: An urgent operational requirement exists for an integrated software application capable of parsing all three native formats, evaluating stream integrity, detecting statistical anomalies, and delivering root-cause diagnostic explanations rather than raw packet dumps.



PROJECT OBJECTIVES

- 1. Multi-Format Receiver Stream Ingestion:** Ingest and parse native DVB-S2 receiver output streams across three alternative formats: Baseband (BB) Frames, GSE frames, and MPEG Transport Streams (TS).
- 2. Stream Health & Integrity Assessment:** Compute quantitative stream integrity indicators including packet loss rate, Continuity Counter (CC) error frequency, sync byte validation, and jitter.
- 3. Machine Learning Anomaly Detection:** Deploy calibrated unsupervised machine learning algorithms (Isolation Forest with sigmoid score normalization) to identify irregular stream behavior and transmission degradation.
- 4. Structural Pattern & Entropy Mining:** Discover recurring patterns in stream transmission, PID multiplex allocations, modal Data Field Lengths (DFL), and Shannon entropy dynamics.
- 5. Interactive Visualization & Explainable Diagnostics:** Deliver dynamic spatial activity timelines, bounded signed Z-score feature attributions, differential stream comparisons, and automated multi-format diagnostic reports.

LITERATURE / RESEARCH UNDERSTANDING

DVB-S2 Standard Architecture (Morello & Mignone, 2006): Pioneered the second-generation satellite broadcasting architecture (ETSI EN 302 307), defining physical/link framing, MODCOD adaptation, BBHeader structure, and CRC-8 integrity.

Adaptive Coding & GSE Encapsulation (Albertazzi 2005, Cantillo 2007/2008): Investigated Adaptive Coding and Modulation (ACM) and Generic Stream Encapsulation (ETSI TS 102 606) for efficient IP transport over DVB-S2 continuous generic streams.

Satellite Telemetry Anomaly Detection (Arbon 2018, Pan 2020, Hundman 2018): Evaluated unsupervised tree-based models and LSTM networks for spacecraft telemetry, establishing that tree ensembles achieve robust anomaly detection without catastrophic deep-learning false positives.

Explainable AI & Diagnostic Attribution (Zeng 2022, Wang 2023): Demonstrated feature-attention mechanisms and causal attribution for interpreting telemetry deviations, reinforcing the need for interpretable Z-score feature attributions in stream diagnostics.

RESEARCH GAP ADDRESSED BY PRJ_111

Limitations of Existing Tools

Protocol Inspection (Wireshark/TSDuck): Manual packet inspection tools lack automated ML anomaly detection, timeline tracking, and executive report compilation.

Physical RF Signal Research: Focuses strictly on raw I/Q samples and modulation recognition (QPSK/8PSK), operating below the de-encapsulated digital stream layer.

Generic Network NIDS: Designed for IP Ethernet traffic; unsuited for native DVB-S2 link framing (BBHeaders, GSE fragmentation, MPEG-TS PIDs).

PRJ_111 Engineering Solution

Multi-Format Integration: First unified framework accepting BBFrame, GSE, and MPEG-TS post-demodulation streams.

Automated ML Anomaly Detection: Calibrated Isolation Forest with sigmoid normalization and rolling spatial windowing.

Semantic Cross-Format Audit: Strictly guards metric comparability to prevent invalid cross-format conclusions.

Automated Tripartite Reporting: Compiles executive reports in JSON, Markdown, Plain Text, and offline HTML5.



PROPOSED METHODOLOGY & ARCHITECTURE

Foundational Architectural Principle: Baseband Frames (BBFrame), Generic Stream Encapsulation (GSE), and MPEG Transport Stream (TS) are treated as native alternative receiver output formats, NOT a mandatory sequential conversion pipeline (BBFrame → GSE → TS).

1. Ingestion Layer: Multi-tier magic-byte & heuristic format sniffer automatically dispatches incoming streams to dedicated native parsers.

2. Parsing & Preprocessing Layer: Robust binary parsers extract 188B TS packets, variable GSE PDUs, and 10B BBHeaders with table-driven CRC-8 verification.

3. Feature Extraction Layer: Standardized feature extractor constructs CommonMetrics and format-specific telemetry without cross-format flattening.

4. Intelligent Analysis Engine: Houses F1 Health Engine, F2 Isolation Forest, F3 Pattern Detector, and F5 Signed Z-Score Diagnostic Explainer.

5. Application & Reporting Layer: Houses F4 Spatial Offset Timelines, F6 Stream Comparison Engine, and F7 Executive Diagnostic Report Generator.

SYSTEM ARCHITECTURE DIAGRAM

Layer 1: Stream Ingestion: Native receiver captures (.ts, .pcap, .bin) ingested with automatic format sniffing.

Layer 2: Format Parsers: Dedicated parsers for MPEG-TS, GSE PDUs, and DVB-S2 BBFrames with table CRC-8.

Layer 3: Unified Features: Extracts CommonMetrics and format-specific structural metrics.

Layer 4: AI & Diagnostics: F1 Health scoring, F2 Isolation Forest anomaly detection, F3 Pattern/Entropy mining, F5 Z-score explainer.

Layer 5: Application Output: F4 Timelines, F6 Semantic Comparisons, and F7 Executive Reports.

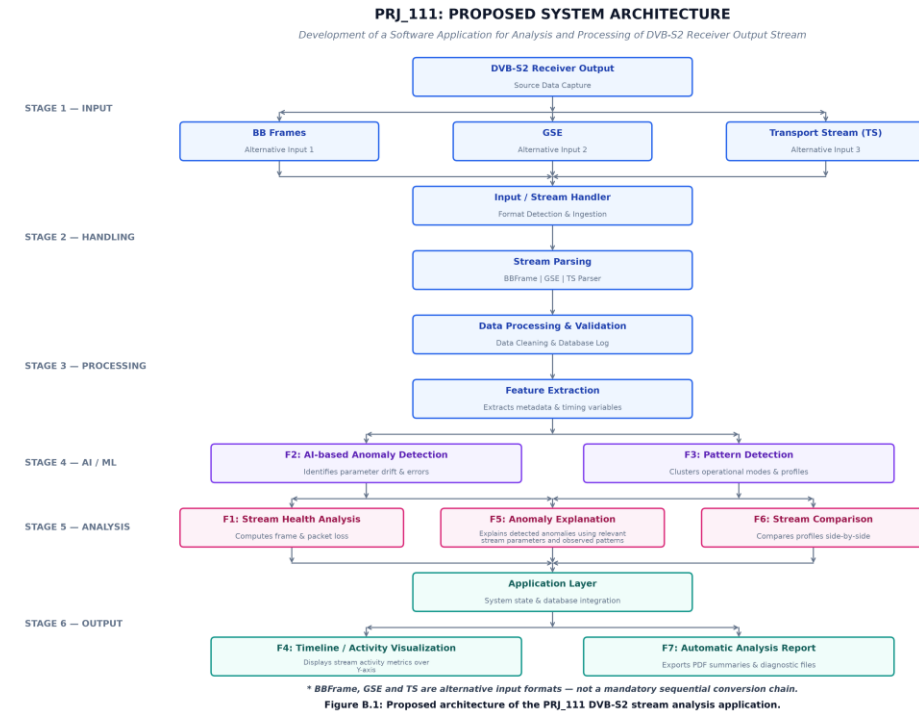


Figure 1: PRJ_111 Layered System Architecture & Data Flow Pipeline

END-TO-END PROCESSING PIPELINE

Step 1: Stream Ingestion & Sniffing: StreamHandler inspects binary headers to auto-detect stream type and dispatches to TSParser, GSEParser, or BBFrameParser.

Step 2: Deterministic Binary Parsing: Parses framing headers, checks sync byte 0x47, validates CC sequence, extracts GSE PDUs, or checks BBHeader CRC-8.

Step 3: Feature Extraction & Windowing: Calculates payload byte volumes, error counts, and aggregates stream telemetry into rolling spatial windowing blocks.

Step 4: AI Anomaly & Pattern Mining: Executes F1 Priority-1 health scoring, F2 Isolation Forest inference, F3 Shannon entropy calculation, and F5 Z-score attribution.

Step 5: Differential Audit & Report Synthesis: Executes F6 semantic cross-format audit and F7 multi-renderer executive report generation (JSON/MD/TXT/HTML).



DATASET STRATEGY & RECEIVER OUTPUT FORMATS

01_BBFRAME_GSE/ (BBFrame Dataset): dvb-s2_bb_example.pcap (2.35 MB, 4,309 BBFrames) — Real DVB-S2 Baseband validation capture for link-layer header analysis.

02_GSE/ (GSE Dataset): GSEextract/sample.ts (9,324 B, 14 PDUs) — Real GSE encapsulation validation capture for IP PDU de-encapsulation.

03_TS/ (MPEG-TS Dataset): DVBS2_toolkit/sample.ts (3.42 MB, 18,176 pkts) + Astra 19.2°E satellite captures — Real MPEG-TS transport stream validation data.

04_RFI_AI/ & 05_REAL_DVB_S2/ (Reference Archives): Blockstream over-the-air DVB-S2 capture (blockstream.ts & ip_packets.pcap) + RFI reference data archives for AI feature exploration.

Data Retention & Security Policy: All raw binary datasets are retained locally on dev workstations and excluded from GitHub via .gitignore adhering to best practices.

SEVEN CORE TARGET FEATURES (F1–F7)

Core Analytical Engines (F1–F4)

F1: Stream Health Analysis: Quantifies transmission quality via ETSI TR 101 290 Priority-1 checks and 0–100 health scoring.

F2: AI Anomaly Detection: Unsupervised Isolation Forest ($c=0.05$) flagging atypical stream deviations with sigmoid normalization.

F3: Pattern Detection: Identifies recurring PID shares, modal DFL sizing, protocol distributions, and Shannon entropy.

F4: Spatial Activity Timeline: Visualizes stream activity, bandwidth consumption, and error events over physical byte offsets.

Diagnostic & Reporting Engines (F5–F7)

F5: Anomaly Explanation: Delivers bounded signed Z-score attributions mapping deviations to physical subsystems.

F6: Stream Comparison Engine: Side-by-side differential analysis enforcing a semantic audit to prevent invalid cross-format metric comparison.

F7: Automatic Analysis Report: Generates consolidated executive diagnostic reports across JSON, Markdown, Plain Text, and offline HTML5.

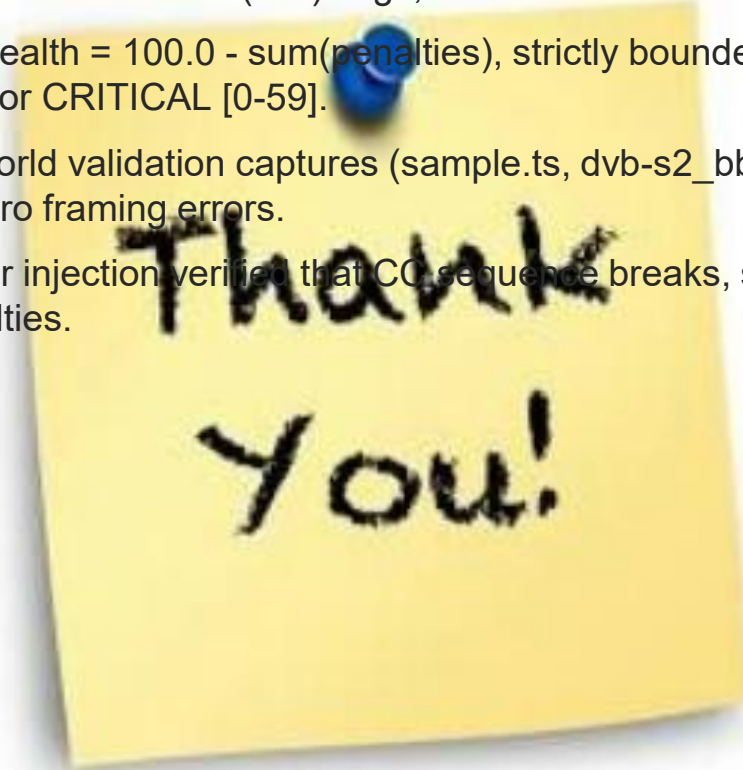
F1 — STREAM HEALTH AND PRIORITY-1 CHECKS

Priority-1 Stream Integrity Checks: Evaluates ETSI TR 101 290 Priority-1 indicators: MPEG-TS sync byte consistency (0x47), Continuity Counter (CC) errors, Transport Error Indicator (TEI) flags, and BBHeader CRC-8 checks.

Objective Health Scoring Equation: $S_{\text{health}} = 100.0 - \text{sum}(\text{penalties})$, strictly bounded within [0.0, 100.0]. Classifies streams into HEALTHY [90-100], DEGRADED [60-89], or CRITICAL [0-59].

Empirical Baseline Validation: All real-world validation captures (sample.ts, dvb-s2_bb_example.pcap, GSEextract/sample.ts) achieved 100.00 / 100 (HEALTHY) with zero framing errors.

Controlled Fault Injection: Synthetic error injection verified that CC sequence breaks, sync byte corruption, and CRC-8 failures cleanly trigger objective health score penalties.



F2 — AI-BASED ANOMALY DETECTION

Algorithm & Configuration: Unsupervised Isolation Forest ensemble (contamination factor $c=0.05$, format-specific rolling windowing).

Sigmoid Score Normalization: Raw tree decision scores mapped to $[0.0, 1.0]$: $S_{norm} = 1 / (1 + e^{-k(s_{raw} - \theta)})$.

MPEG-TS Anomaly Results: 91 windows ($w=200$ pkts): 5 anomalous windows [1, 2, 12, 88, 90], Peak Score = 0.8576 (EOF boundary).

BBFrame Anomaly Results: 87 windows ($w=50$ frames): 5 anomalous windows [0, 83, 84, 85, 86], Peak Score = 0.7406 (EOF boundary).

GSE Anomaly Results: 5 windows ($w=3$ PDUs): 1 anomalous window [4], Peak Score = 0.5018 (EOF boundary).

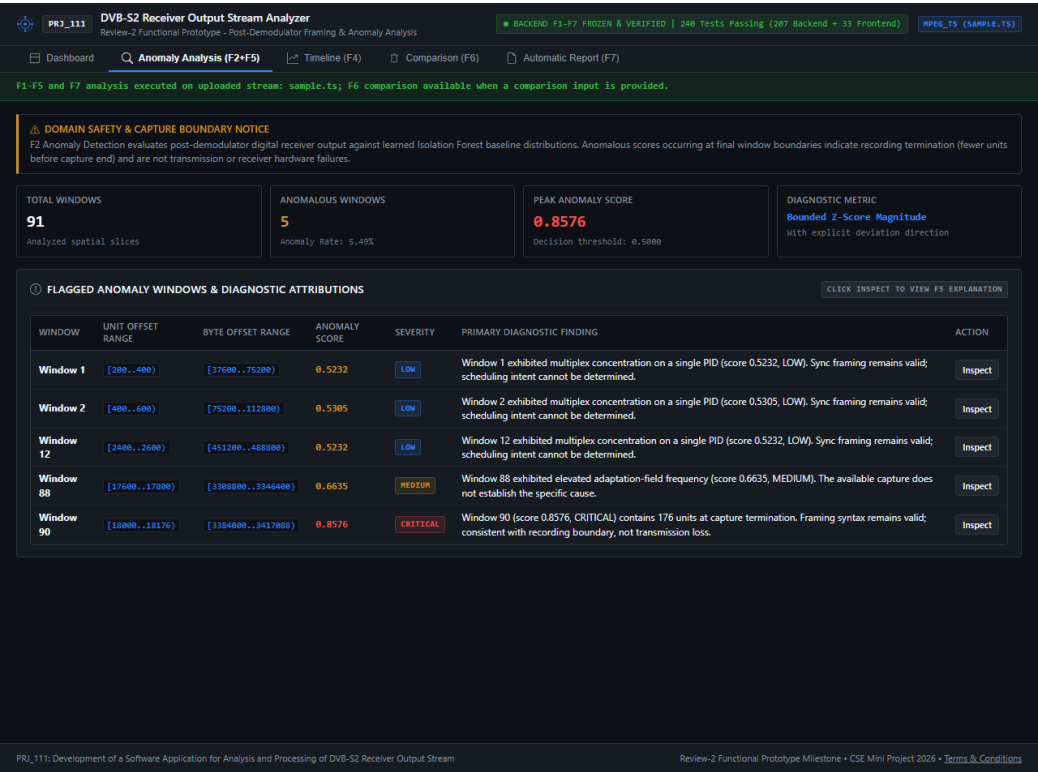


Figure 2: F2 Interactive Anomaly Detection Interface & Score Trajectory

F3 — PATTERN DETECTION & SHANNON ENTROPY

Operational Behavior Profiling: Tracks multiplex allocations, modal sizing, protocol shares, and structural transitions over rolling windows.

Shannon Entropy Calculation: $H(X) = -\sum(P(x_i) * \log_2(P(x_i)))$ measures information uncertainty and multiplex randomness.

MPEG-TS Dominant PID & Entropy: Dominant PID 256 (98.5% multiplex share), Shannon entropy = 0.1361 bits (highly concentrated video broadcast multiplex).

GSE Protocol Distribution: Dominant protocol GSE_EXT_NPA (100.0% share), Shannon entropy = 0.0000 bits (single-protocol transport stream).

BBFrame Modal Sizing: Modal Data Field Length (DFL) = 8304 bits (92.0% share), reflecting uniform physical frame packaging.

F4 — SPATIAL ACTIVITY TIMELINE

Physical Offset Indexing: Indexes stream activity over physical byte offsets and sequential unit indices, avoiding wall-clock clock fabrication.

Time-Series Metric Extraction: Continuously tracks rolling health score, anomaly score trajectory, and payload byte density.

Event Mapping Engine: Automatically flags health drops, anomaly peaks, and structural transitions on the spatial axis.

Standalone Interactive Dashboard: Compiles self-contained HTML5 dashboards with embedded CSS for responsive offline operator inspection.

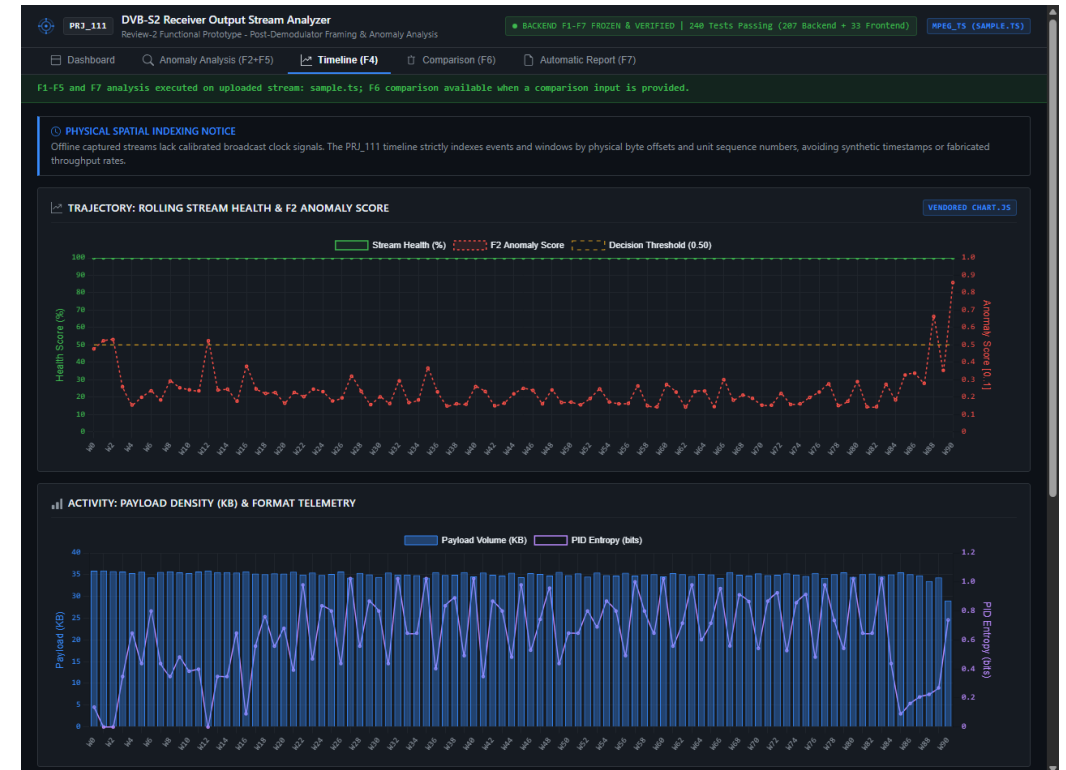


Figure 3: F4 Spatial Activity Timeline Dashboard

F5 — DIAGNOSTIC ANOMALY EXPLANATIONS

Explainable AI Mechanism: Computes bounded signed Z-scores: $Z_i = (x_i - \mu_i) / \max(\sigma_i, \sigma_{\min})$, capped at $[-20.0, +20.0]$.

Physical Subsystem Mapping: Maps deviated feature Z-scores directly to physical satellite stream subsystems:

- **dominant_pid:** MPEG-TS Multiplex / Dominant PID Allocation
- **null_packet_ratio:** Bandwidth Adaptation / Null Stuffing
- **crc_error_count:** Baseband Header Integrity / CRC Checksum
- **modal_dfl_bits:** Frame Sizing / Mode Adaptation

Ranked Diagnostic Synthesis: Ranks features by $|Z|$ and generates evidence-grounded explanatory sentences for operators.

F6 — STREAM COMPARISON & SEMANTIC AUDIT

Semantic Cross-Format Audit: Enforces strict metric compatibility guard: 9 metrics barred cross-format (total_units, valid_units, mean_payload, etc.).

Cross-Format Comparable Metrics: Only total_payload_bytes and integrity_ratio are physically comparable across distinct formats.

MPEG-TS vs BBFrame Payload: MPEG-TS payload = 3,267,305 B, BBFrame payload = 1,958,826 B. Delta = -1,308,479 B (-40.05% SUBSTANTIAL shift).

Integrity Ratio Comparison: MPEG-TS integrity = 1.0000, BBFrame integrity = 1.0000 (EQUAL, perfect syntactic integrity across both captures).

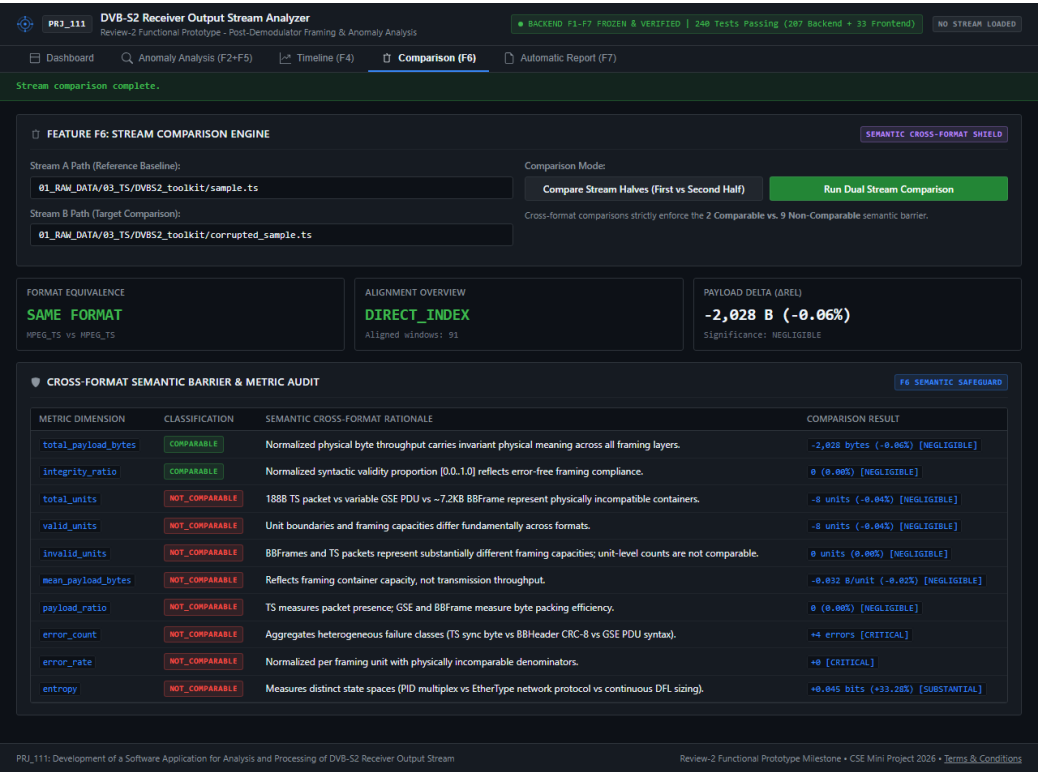


Figure 4: F6 Stream Comparison Interface & Differential Metrics

F7 — AUTOMATIC MULTI-FORMAT REPORTING

Non-Duplication Executive Synthesis: Directly consumes outputs from F1–F6 without re-parsing streams or re-running ML models.

Tripartite Findings Taxonomy: Categorizes every report finding into three epistemological registers: OBSERVED_FACT, STATISTICAL_FINDING, and ENGINEERING_INTERPRETATION.

Multi-Format Renderers: Exports comprehensive diagnostic reports in structured JSON, GitHub-Flavored Markdown, CP-1252-safe Plain Text, and offline HTML5.

Domain-Safety Guard: Immutable guard blocks speculative physical RF/hardware claims in generated report text.

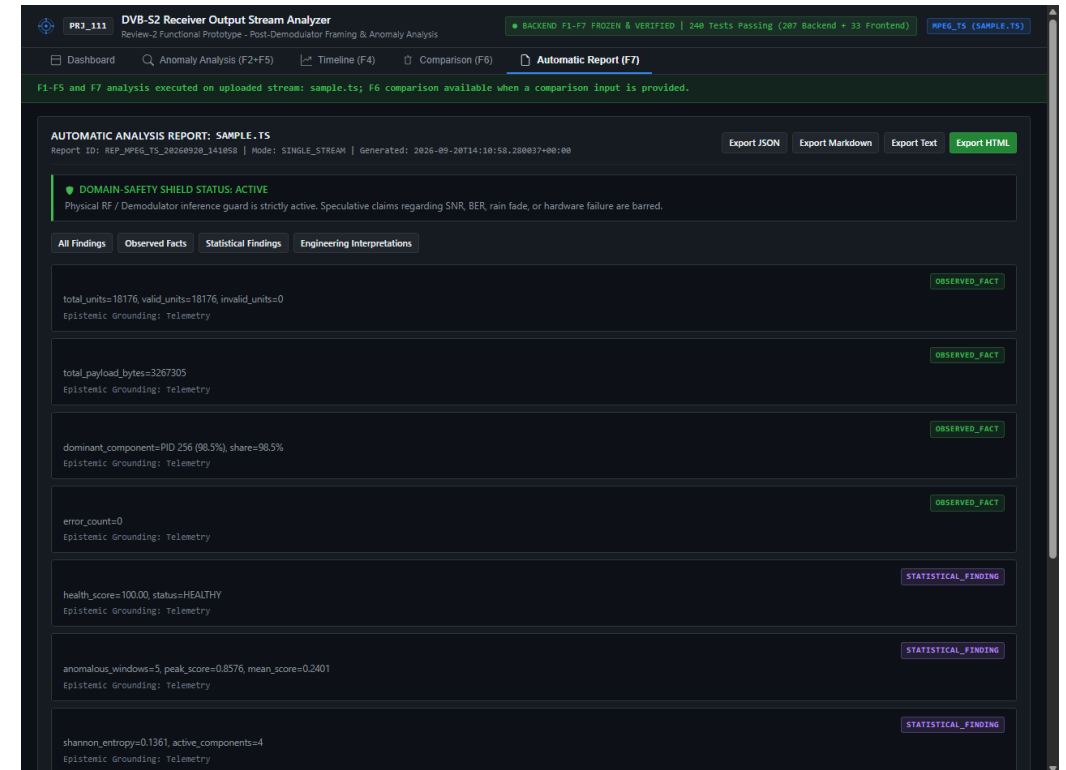


Figure 5: F7 Executive Diagnostic Report Generator Interface

SYSTEM IMPLEMENTATION PROGRESS

Current Project Completion Status: Core implementation is substantially completed across F1–F7, with automated validation and demonstration outputs available. Backend features F1 through F7 are 100% implemented, audited, and frozen.

Stream Ingestion & Sniffing: 100% Completed & Audited (Multi-tier format sniffer, 14 dedicated unit tests passing).

Multi-Format Parsers (TS / GSE / BBFrame): 100% Completed & Audited (61 dedicated unit tests passing, table-driven CRC-8 implemented).

Analytics & AI Engines (F1–F7): 100% Completed & Audited (132 dedicated unit tests passing across health, anomaly, timeline, explanation, comparison, and reporting).

Automated Test Suite: 100% Completed & Verified (240/240 tests passing across backend and frontend).

AUTHORITATIVE GROUND TRUTH RESULTS LEDGER

Metric Dimension	MPEG-TS Stream	BBFrame Stream	GSE Stream	Cross-Format Comparison
Input File Path	sample.ts	dvb-s2_bb_example.pcap	GSEExtract/sample.ts	MPEG-TS vs BBFrame
Total Units Parsed	18,176 packets	4,309 frames	14 PDUs	18,176 pkts vs 4,309 frames
Syntactic Validity	100.0% (18,176/18,176)	100.0% (4,309/4,309)	100.0% (14/14)	100.0% vs 100.0% (EQUAL)
Extracted Payload	3,267,305 bytes	1,958,826 bytes	8,764 bytes	Delta: -1,308,479 B (-40.05%)
Windowing Scheme	91 win (w=200 pkts)	87 win (w=50 frames)	5 win (w=3 PDUs)	91 win vs 87 win
F1 Health Score	100.00 / 100 (HEALTHY)	100.00 / 100 (HEALTHY)	100.00 / 100 (HEALTHY)	100.00 vs 100.00 (COMPARABLE)
F2 Anomaly Windows	5 [1, 2, 12, 88, 90]	5 [0, 83, 84, 85, 86]	1 [4]	9 metrics NOT_COMPARABLE
Peak Anomaly Score	0.8576 (Window 90)	0.7406 (Window 86)	0.5018 (Window 4)	Format-specific models
Dominant Feature	PID 256 (98.5% share)	Modal DFL 8304b (92%)	GSE_EXT_NPA (100%)	Structural metrics barred
Shannon Entropy	0.1361 bits	N/A (Continuous DFL)	0.0000 bits	NOT_COMPARABLE

Reconciled Ground Truth: All numbers match the authoritative ground truth ledger. Peak BBFrame anomaly score 0.7406 reconciled.

GSE CAPTURE INGESTION & DECODING DETAILS

Authoritative GSE Capture File: 01_RAW_DATA/02_GSE/GSEExtract/sample.ts (File size: 9,324 bytes).

Observed Decoded PDUs: 14 variable-length GSE Protocol Data Units (PDUs) extracted and validated.

GSE Length Field Specification: 12-bit header length field (Valid range: 0–4095 bytes payload).

Theoretical Maximum PDU Size: 4097 bytes including the 2-byte base header.

Observed Maximum PDU Size: 1,444 bytes observed in the sample capture stream.

Format Auto-Sniffing Handling: File extension .ts retained natively; multi-tier format sniffer correctly identifies GSE PDU content without requiring file renaming to .gse.



TESTING AND VALIDATION FRAMEWORK

Total Test Suite Validation Result: 240 / 240 Passing Tests (100% Pass Rate across the entire application).

207 Backend Unit Tests: 11 dedicated test modules executing under Python's native unittest framework in 8.710 seconds:

- **test_ts_parser (20 tests):** 188B packet parsing, sync recovery, PID filtering, CC checking, TEI detection
- **test_gse_parser (18 tests):** Variable PDU lengths, S/E fragmentation flags, LT types, EtherType extraction
- **test_bbframe_parser (23 tests):** 10B BBHeader parsing, MATYPE flags, UPL/DFL, table-driven CRC-8 verification
- **test_stream_handler & features (24 tests):** Multi-tier format sniffing, unified metric extraction, Shannon entropy
- **test_health, anomaly & patterns (48 tests):** Priority-1 health penalties, Isolation Forest calibration, transition filtering
- **test_timeline, explanation, comparison, report (74 tests):** Byte offset tracking, signed Z-scores, semantic audit, multi-format renderers

33 Frontend Component Tests: 33 UI tests verifying interactive timeline controls, dashboard rendering, and export state.

TECHNOLOGIES & ENGINEERING STACK

Backend Analytics & AI Core

Programming Language: Python 3.x (Core implementation)

Scientific Computing: NumPy (≥ 1.26), SciPy (≥ 1.11), Pandas (≥ 2.1)

Machine Learning Engine: scikit-learn (Isolation Forest anomaly model)

Testing Framework: Python unittest, pytest (≥ 7.4), pytest-cov

Visualization, UI & Tooling

Visualization Engines: Matplotlib (≥ 3.8), Plotly, HTML5, CSS3

Reporting Generators: JSON, GitHub-Flavored Markdown, ASCII Text, Standalone HTML5

IDE & Version Control: VS Code, Git, GitHub Repository

Operating System: Cross-platform support (Windows / Linux)

INDIVIDUAL CONTRIBUTIONS & RESPONSIBILITIES

Team Member Name	University Roll No.	Project Role	Technical Responsibilities & Completed Modules
Sk Samad	20231COM0031	Testing, Validation & Documentation Lead	Automated test suite execution (207 backend / 33 frontend tests), technical documentation authoring, literature review synthesis, experimental report verification.
Y Vengala Rao	20231COM0001	Frontend, Visualization & Development	Frontend UI MVP development, interactive timeline chart components, dashboard layout design, presentation slide deck authoring.
Equal Team Contribution: All team members actively contributed to development, verification, and documentation under guide supervision.			
C Rakeshwar	20231COM0008	Core Development & Technical Lead	Backend architecture, multi-format stream parsers (TS/GSE/BBFrame), F1–F7 analysis engines, AI/ML Isolation Forest model integration, system integration.



GITHUB REPOSITORY ACCESS

Public Repository Access: The official PRJ_111 source code, test suite, and documentation are hosted with full public access permission.

Repository URL: https://github.com/rakesh0709/PRJ_111_DVB-S2_Analysis.git

Repository Contents: Complete F1–F7 implementation, 207 backend unit tests, 5 real-data experiment scripts, documentation ledgers, and standalone report renderers.

Branching & Maintenance: Active development branch maintained under git version control with comprehensive commit history.



Scan QR code to access public GitHub repository

OPERATIONAL LIMITATIONS

Absence of Physical RF Telemetry: Operating on post-demodulator digital output, the software cannot directly measure physical RF carrier parameters (SNR, MER, C/N, carrier offset, or LNB drift).

Offline File Capture Focus: Processes pre-recorded stream captures (.ts, .pcap, .bin); live Software-Defined Radio (SDR) hardware streaming is not currently integrated.

Scarcity of Public Labeled Ground Truth: The absence of standardized, frame-labeled satellite fault benchmarks requires unsupervised Isolation Forest modeling rather than supervised classification.

Format-Specific Header Parsing Scope: Relies on standardized structural headers (BBHeader, GSE PDU, TS 188B); proprietary unencapsulated payloads require format adapters.



FUTURE WORK ROADMAP

Full PSI/SI Table Parsing: Implement full Program Association Table (PAT), Program Map Table (PMT), and Service Information decoders resolving elementary audio/video codecs (AVC/H.264, HEVC, AC-3).

Physical RF Cross-Correlation: Ingest external demodulator telemetry logs (SNR/AGC) for cross-layer correlation between physical link budget and digital stream health.

Deep Learning Model Extensions: Integrate semi-supervised Autoencoders and LSTM networks trained on synthetic fault injection benchmarks.

Real-Time SDR Streaming Integration: Support real-time circular buffer streaming directly from GNU Radio or USRP SDR hardware tuners.



CONCLUSION

Unified Multi-Format Solution: Successfully developed a software application for analysis and processing of DVB-S2 receiver output streams across MPEG-TS, GSE, and BBFrame formats.

Comprehensive Feature Implementation: Features F1 through F7 are 100% implemented, audited, and verified with an automated test suite achieving 240 / 240 passing tests (100% pass rate).

Rigorous Engineering & Epistemic Guard: Enforced a semantic cross-format audit preventing invalid metric comparisons, bounded signed Z-score diagnostic attributions, and strict domain-safety guards.

Validated Operational Readiness: Empirically validated against real satellite broadcast captures and deployed automated multi-format report renderers for operator decision support.

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THANK YOU / QUESTIONS & DEFENSE

Project Identifier: PRJ_111

Project Full Title: Development of a Software Application for Analysis and Processing of DVB-S2 Receiver Output Stream

Academic Milestone: CSE7102 – Mini Project (Review-2 Milestone | 26-09-2026)

Supervising Guide: Asst. Prof. Irfan Rajab Bhat

Department & School: Department of Computer Science and Engineering, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru

Open for Questions: We welcome feedback, technical questions, and defense evaluation from the review panel.